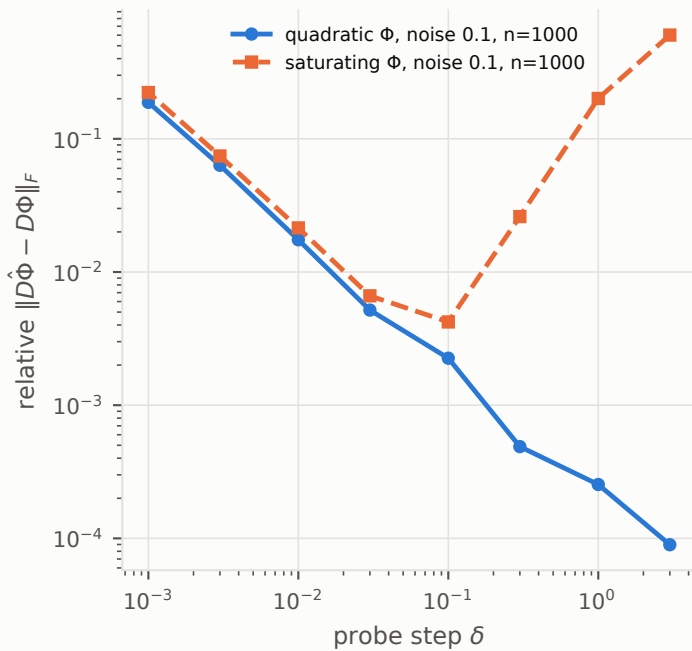
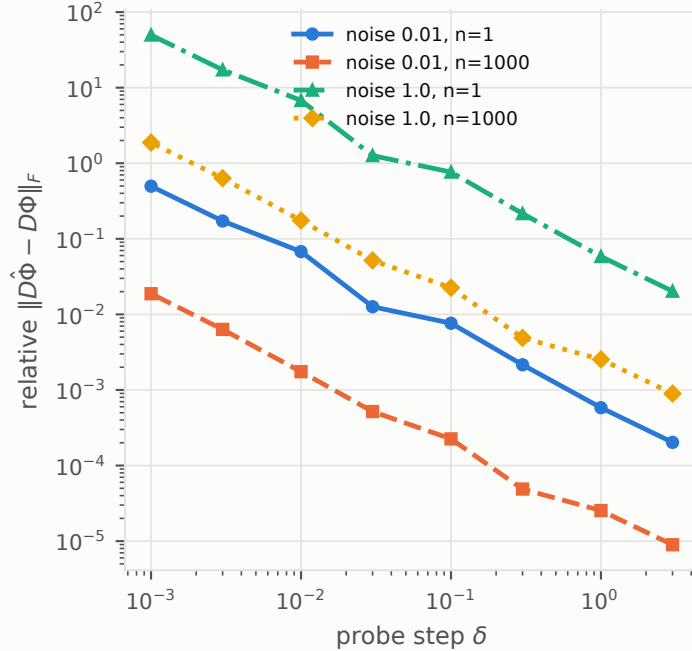
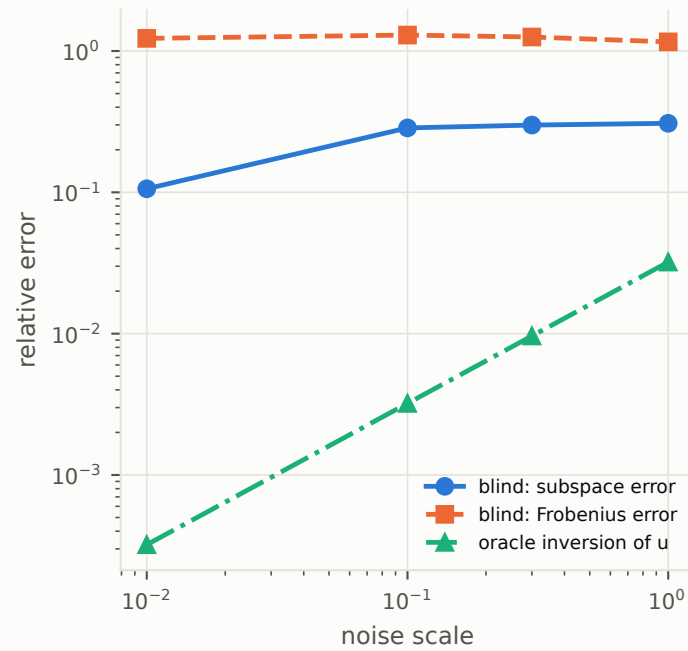
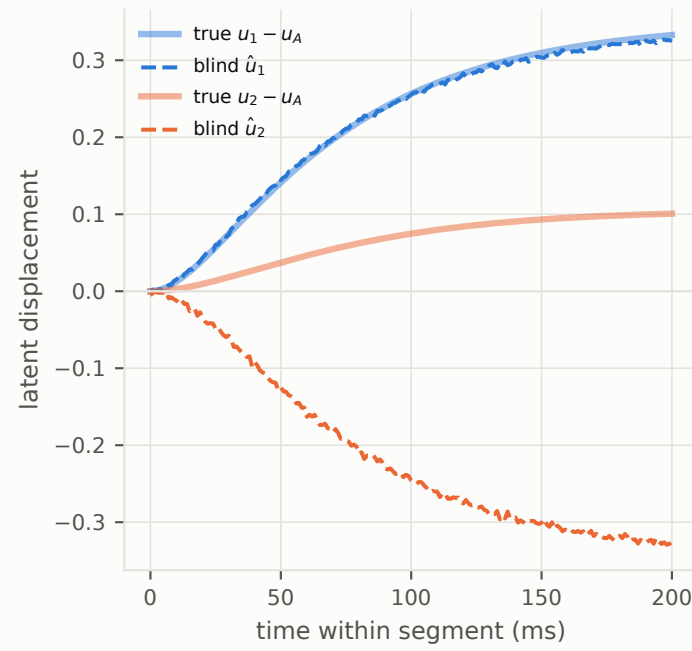


Probe Jacobian: bias appears only with curvature**Noise and averaging, quadratic Φ** **What each estimator can reach****Blind inversion recovers a rotated latent**

Left: the probe estimator trades truncation bias (large δ) against observation noise (small δ); the best step over this sweep is $\delta = 0.03$. Centre: the blind estimator pins the tangent subspace but not the Jacobian. Right: the blind latent is a linear reparameterisation of the true one, which is exactly what an acoustic-only method can identify.